

Question 01

Given, $\beta = 2$, $m = 4$ and $e \in [2, 6]$

① General form,

$$F = \pm (0.1d_2d_3d_4)_2 \times 2^e$$

$$\begin{aligned}\text{Maximum number, } F &= (0.1111)_2 \times 2^6 \\ &= 0.9375 \times 2^6 \\ &= 60\end{aligned}$$

$$\begin{aligned}\text{Minimum number, } F &= -(0.1111)_2 \times 2^6 \\ &= -60\end{aligned}$$

Normalized form,

$$F = \pm (1.d_1d_2d_3d_4)_2 \times 2^e$$

$$\begin{aligned}\text{Maximum number, } F &= (1.1111)_2 \times 2^6 \\ &= 1.9375 \times 2^6 \\ &= 124\end{aligned}$$

$$\begin{aligned}\text{Minimum number, } F &= - (1.1111)_2 \times 2^6 \\ &= -124\end{aligned}$$

Denormalized form,

$$F = \pm (0.1d_1d_2d_3d_4)_2 \times 2^e$$

$$\begin{aligned}\text{Maximum number, } F &= (0.11111)_2 \times 2^6 \\ &= 0.96875 \times 2^6 \\ &= 62\end{aligned}$$

$$\begin{aligned}\text{Minimum number, } F &= -(0.11111)_2 \times 2^6 \\ &= -62\end{aligned}$$

⑥ Non-negative minimum numbers,

$$\begin{aligned}\text{General form, } F &= (0.1000)_2 \times 2^{-2} \\ &= 0.125\end{aligned}$$

$$\begin{aligned}\text{Normalized form, } F &= (1.0000)_2 \times 2^{-2} \\ &= 0.25\end{aligned}$$

$$\begin{aligned}\text{Denormalized form, } F &= (0.10000)_2 \times 2^{-2} \\ &= 0.125\end{aligned}$$

$$\textcircled{2} \quad e[-2, 6] = e[-2, -1, 0, 1, 2, 3, 4, 5, 6]$$

Total exponent 9.

Total possible combinations,

$$\begin{aligned} \text{General form} &= 2^3 \times 9 \times 2 \\ &= 144 \end{aligned}$$

$$\begin{aligned} \text{Normalized form} &= 2^4 \times 9 \times 2 \\ &= 288 \end{aligned}$$

$$\begin{aligned} \text{Denormalized form} &= 2^4 \times 9 \times 2 \\ &= 288 \end{aligned}$$

Question 02

$$\textcircled{a} \quad \text{Given, } x = (5.625)_{10} = (5)_{10} + (.625)_{10}$$

$$\begin{array}{r} 2 \overline{) 5} \\ 2 \overline{) 2-1} \\ 2 \overline{) 1-0} \\ \hline 0-1 \end{array}$$

$$(5)_{10} = (101)_2$$

$$0.625 \times 2 = 1.25$$

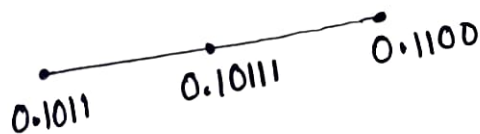
$$0.25 \times 2 = 0.5$$

$$0.5 \times 2 = 1.0$$

$$(0.625)_{10} = (0.101)_2$$

$$\therefore x = (5.625)_{10} = (101.101)_2$$

$$\begin{aligned} \textcircled{b} \quad x &= (101.101)_2 \\ &= (0.101101)_2 \times 2^3 \\ &= (0.1011)_2 \times 2^3 \end{aligned}$$



$$f(x) = (0.1011)_2 \times 2^3$$

$$\begin{aligned} \textcircled{c} \quad f(x) &= (0.1011)_2 \times 2^3 \\ &= \cancel{0.8} 0.6875 \times 2^3 \\ &= 5.5 \end{aligned}$$

$$\begin{aligned} \text{Rounding error, } \delta &= \frac{|f(x) - x|}{|x|} = \frac{|5.5 - 5.625|}{|5.625|} \\ &= 0.02222 \end{aligned}$$

$$\text{Machine epsilon, } \epsilon = \frac{1}{2} \beta^{-m}$$

$$= \frac{1}{2} \cdot \cancel{10^{-3}} \cdot 2^3$$

$$= \cancel{0.0005} \cdot 2^3$$

Question 03

① $x^2 - 60x + 1 = 0$

$$\Rightarrow x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$= \frac{-60 \pm \sqrt{(-60)^2 - 4 \cdot 1 \cdot 1}}{2 \cdot 1}$$

$$\therefore x_1 = 30 + \sqrt{899} = 30 + 29.9833287 = 59.9833287 = 59.9833 \text{ (6-sig fig)}$$

$$f(x_1) = 30 + 29.9833 = 59.9833 \text{ (6-sig fig)}$$

$$\delta_1 = \frac{|f(x_1) - x_1|}{|x_1|} \times 100\%$$

$$= \frac{|59.9833 - 59.9833287|}{|59.9833287|} \times 100\%$$

$$= 0.00004784662776\%$$

$$\therefore x_2 = 30 - \sqrt{899} = 30 - 29.9833287 = 0.01667129887 = 0.0166713 \text{ (6-sig fig)}$$

$$f(x_2) = 30 - 29.9833 = 0.0167 \text{ (4-sig fig)}$$

$$\delta_2 = \frac{|f(x_2) - x_2|}{|x_2|} \times 100\%$$

$$= \frac{|0.0167 - 0.01667129887|}{|0.01667129887|} \times 100\%$$

$$= 0.1721589315\%$$

So, the loss of significance problem occurs at the time of finding the 2nd root. Since, the 2nd root x_2 involves subtraction of two nearly equal numbers, it causes a loss of precision due to significant digits being lost.

⑥ From a,

$$x_1 = 59.9833 \text{ (6-sig fig)}$$

$$x_2 = 0.0166713 \text{ (6-sig fig)}$$

Now,

$$x_1 \cdot x_2 = \frac{c}{a}$$

$$\Rightarrow x_2 = \frac{c}{a \times x_1}$$

$$\Rightarrow x_2 = \frac{1}{1 \times 59.9833}$$

$$\Rightarrow x_2 = 0.01667130685$$

$$\therefore x_2 = 0.0166713 \text{ (6-sig fig)}$$

$\therefore$ Loss of significance does not occur.